

2024 Robotics Project II

Grade

- Code (40%)
 - One-click execution, please set the parameters and measurement data in advance (if you need to install additional actions or extra software or libraries, please specify in the report).
 - Please label the annotation
 - Need to draw 3D trajectory of end point position and direction.
 - Plot the change of six-axis variables (θ_1 、 θ_2 、 θ_3 、 θ_4 、 θ_5 、 θ_6), velocity, acceleration.
 - Plot the change in end point position (x,y,z), velocity, and acceleration.
 - Bonus : Make an animation of a six-axis arm.
- Report (60%)
 - Interface description (development platform, how to run ...)
 - Program architecture description (program flow, core code description ...)
 - Mathematical derivation process
 - Results of trajectory planning
 - Bonus : Discuss the advantages and disadvantages of the two types of trajectory planning, and how to deal with singularities.

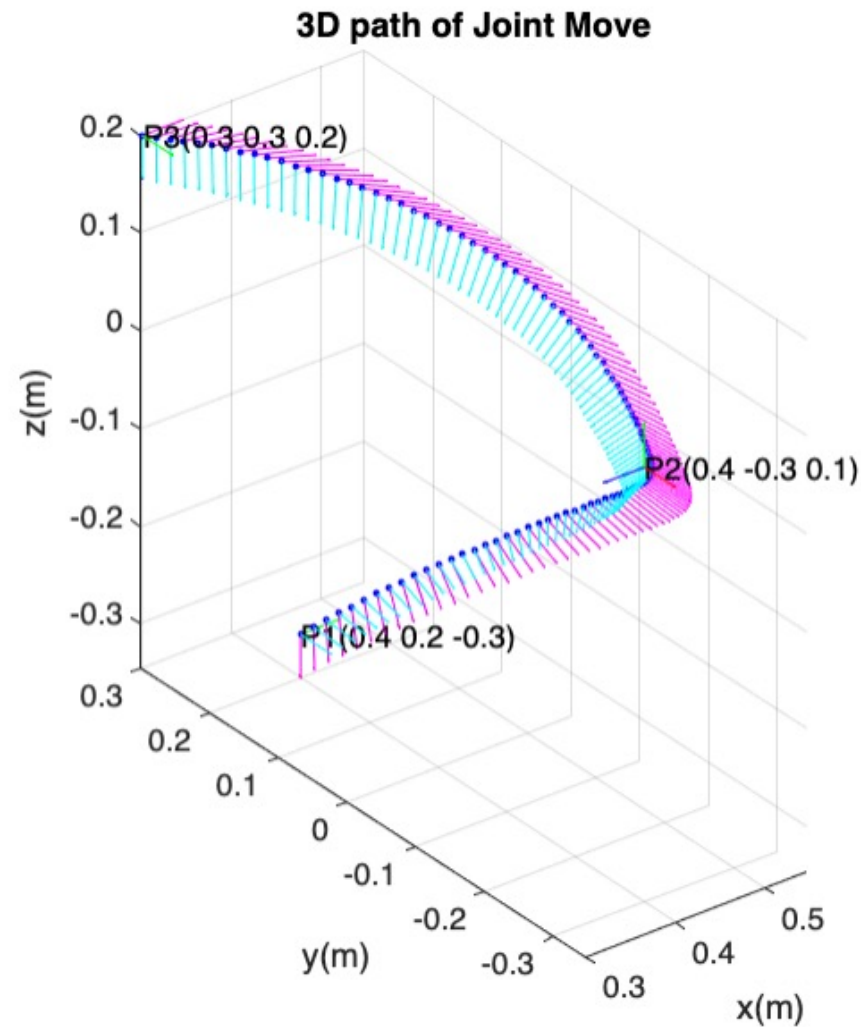
Note

- Please submit the result report as an electronic file (Robotics_Project2_Student-ID.pdf).
- Please compress the complete project (Code + report) into "Robotics_Project2_Student-ID.zip" .
- The project can be done in any programming language, e.g. MATLAB, Python, C/C++ , C#, Java ... etc. Please explain how to do it in the result report. Please explain how to do this in the results report.
- The mathematical description of the result report can be scanned by hand, and it is preferable to use an equation tool to present the mathematical calculations.

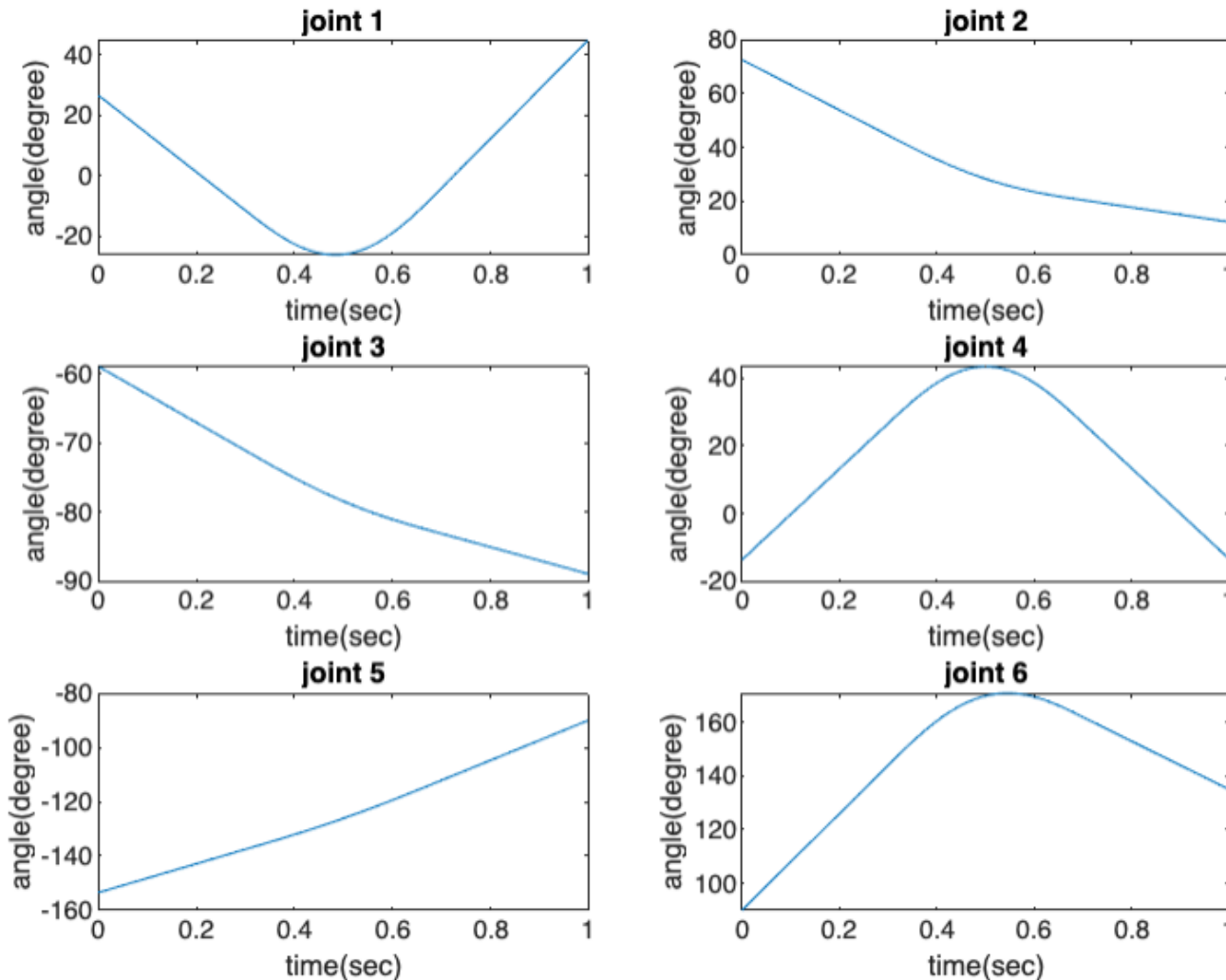
Note II

- Please do not plagiarize.
- Please make sure the inverse kinematics can be executed correctly, or you can use an open-source program.
- When planning the trajectory, please pay attention to whether the unit is correct or not, the unit of length is millimeter (mm).
- This project focuses on the algorithm of trajectory planning. It is accepted to use any open-source tools (need to attach the source).

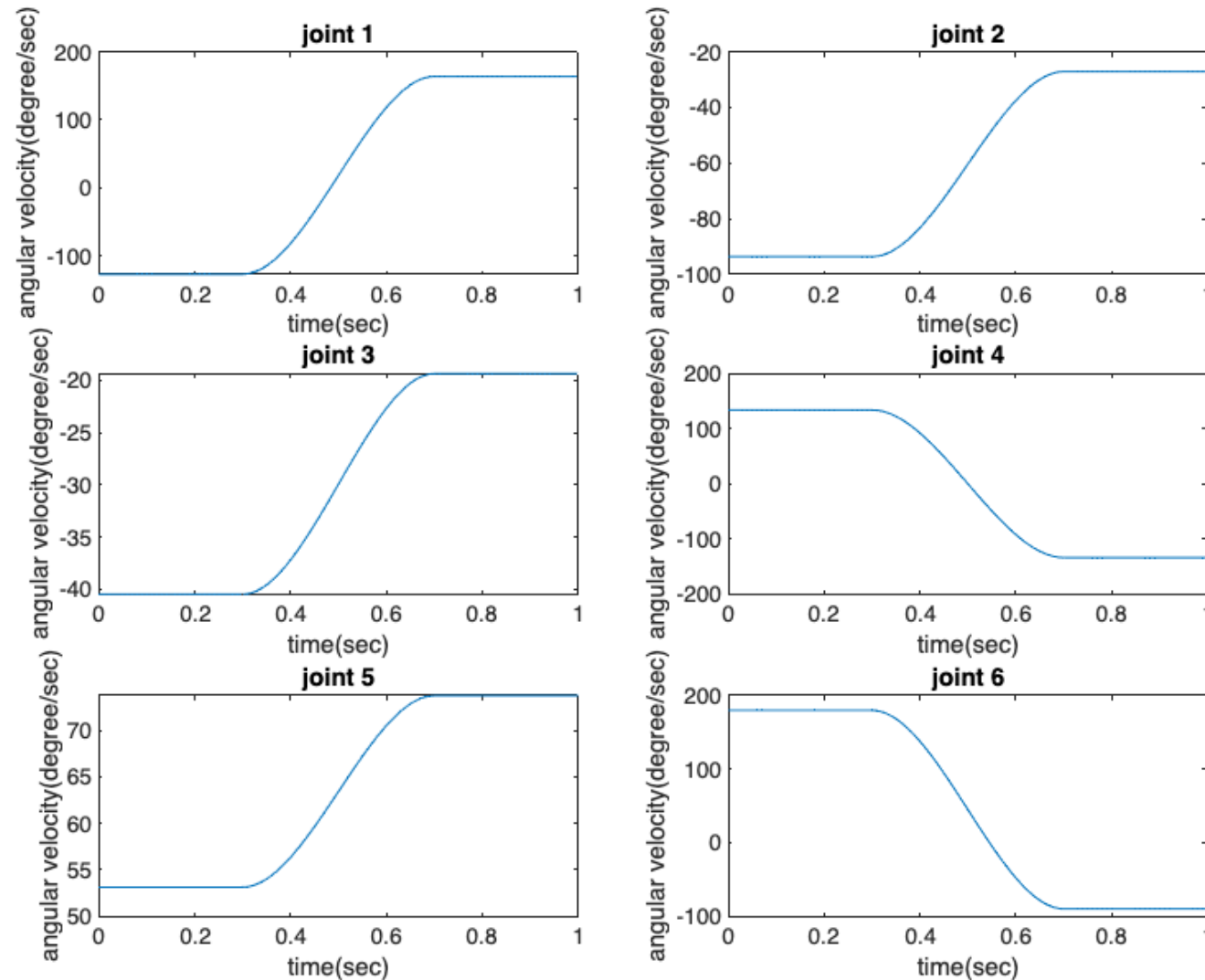
Example : (a) Joint move–3D plot



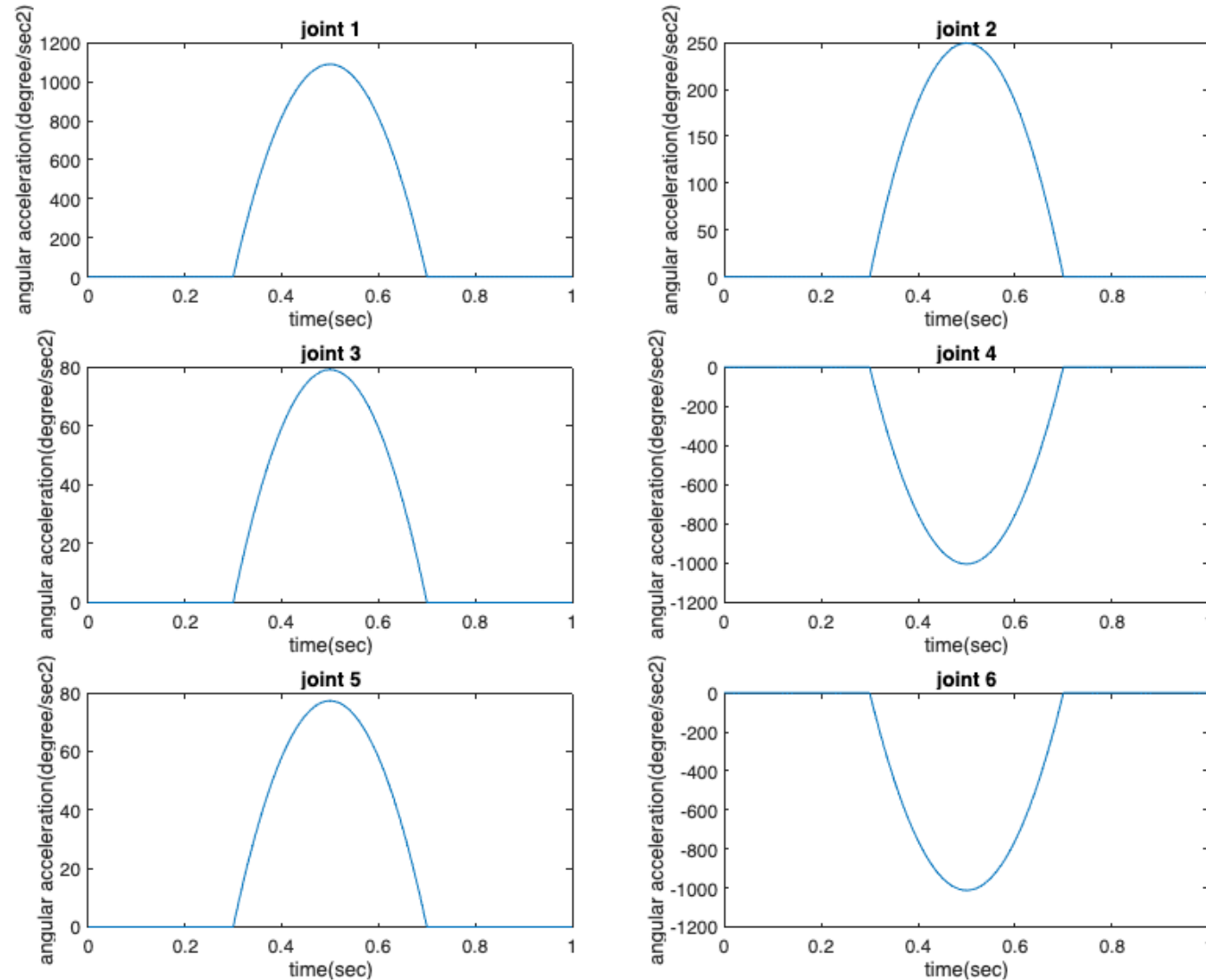
Example : (a) Joint move – angle



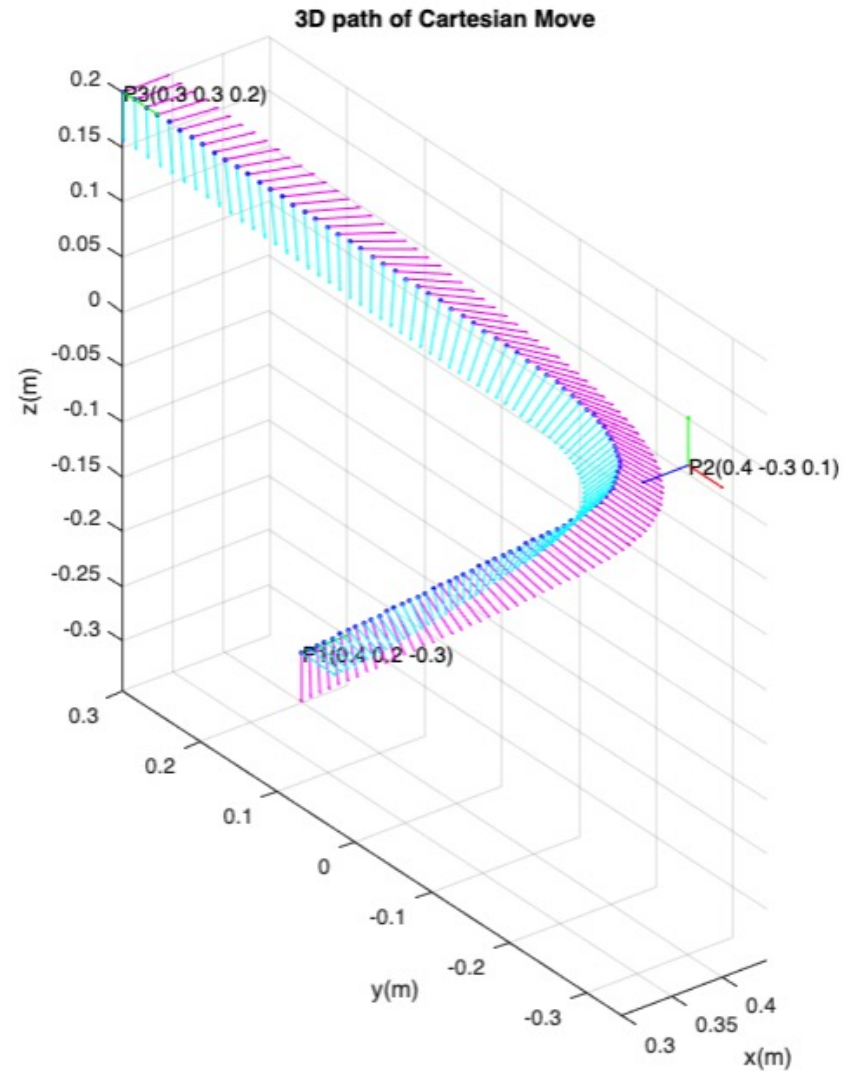
Example : (a) Joint move–angular velocity



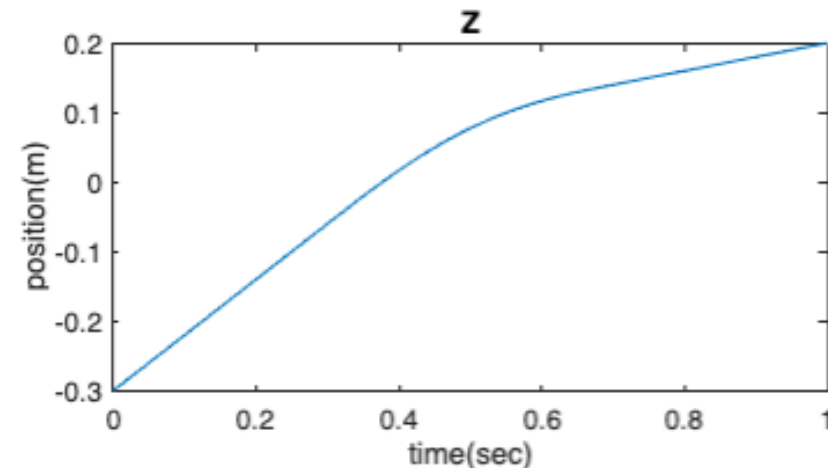
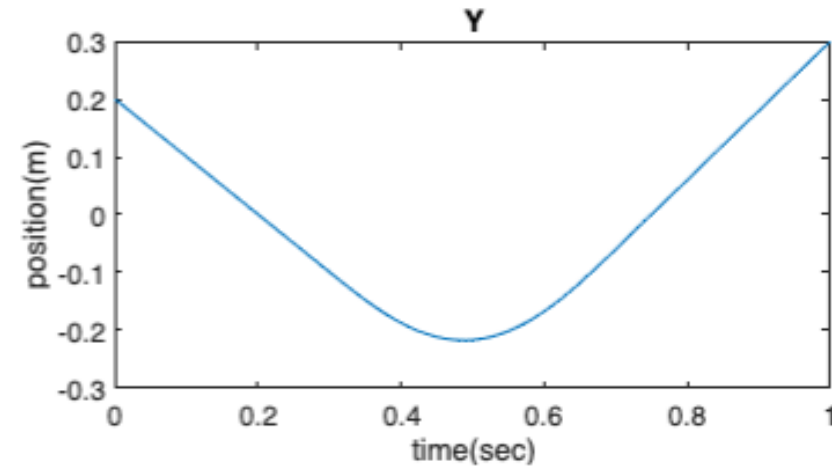
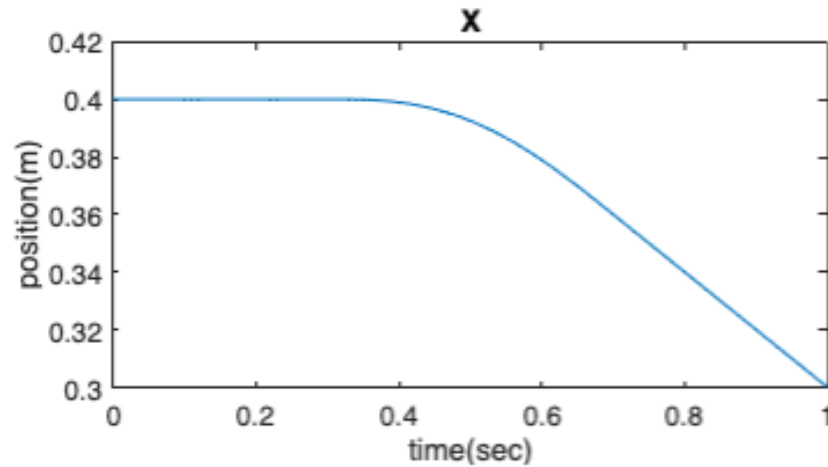
Example : (a) Joint move – angular acceleration



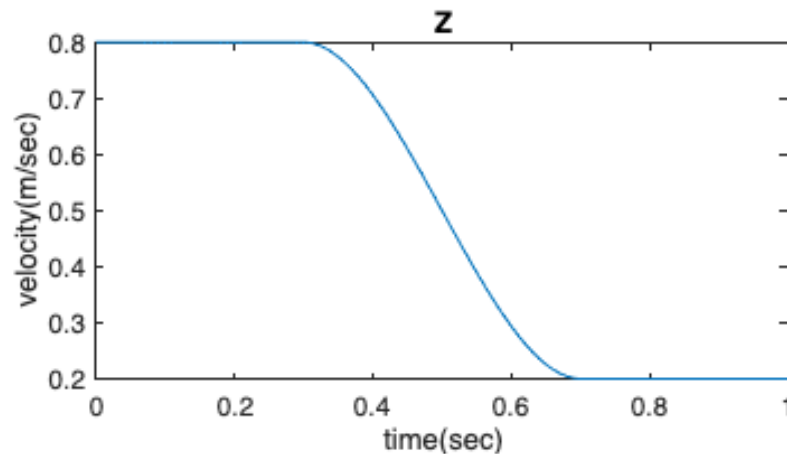
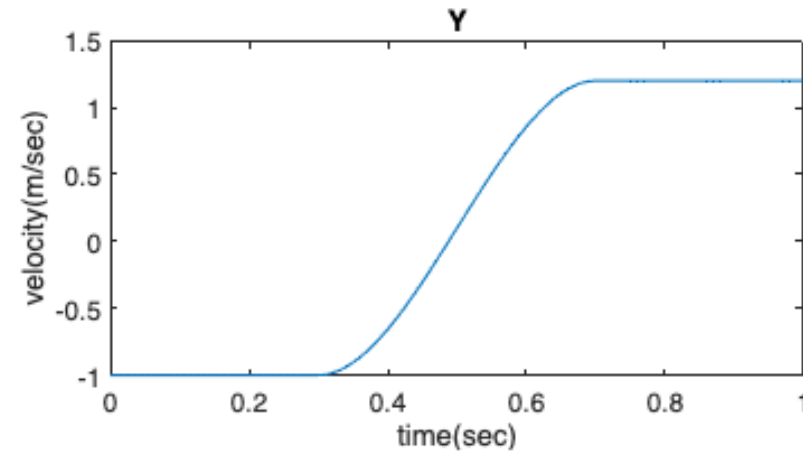
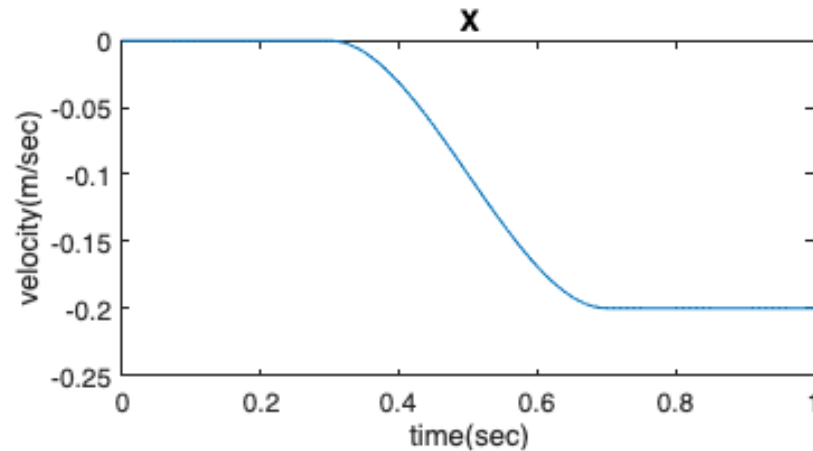
Example : (b) Cartesian move – 3D plot



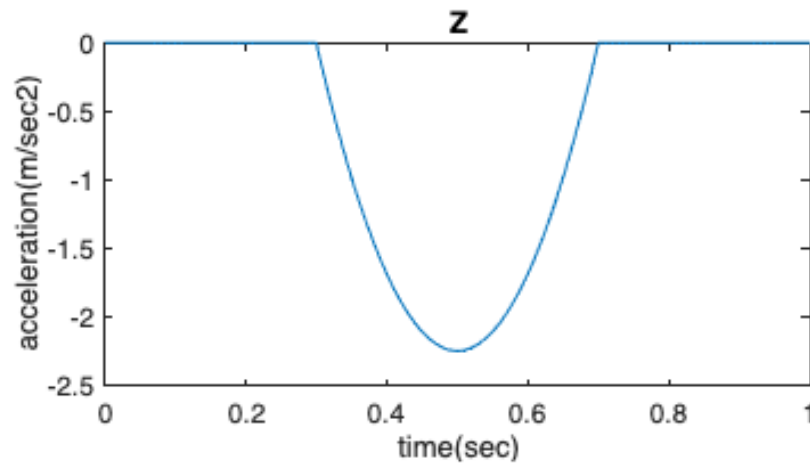
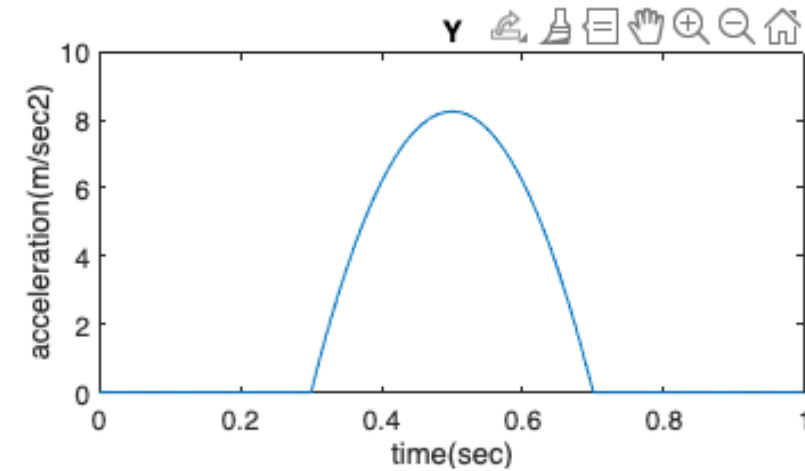
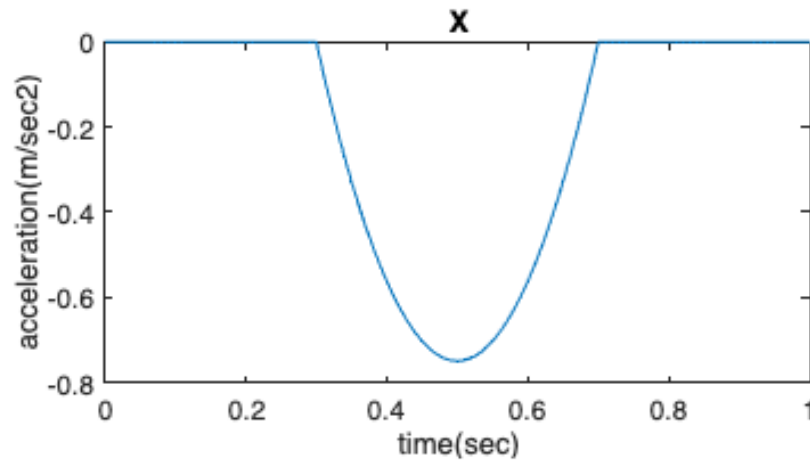
Example : (b) Cartesian move - position



Example : (b) Cartesian move - velocity

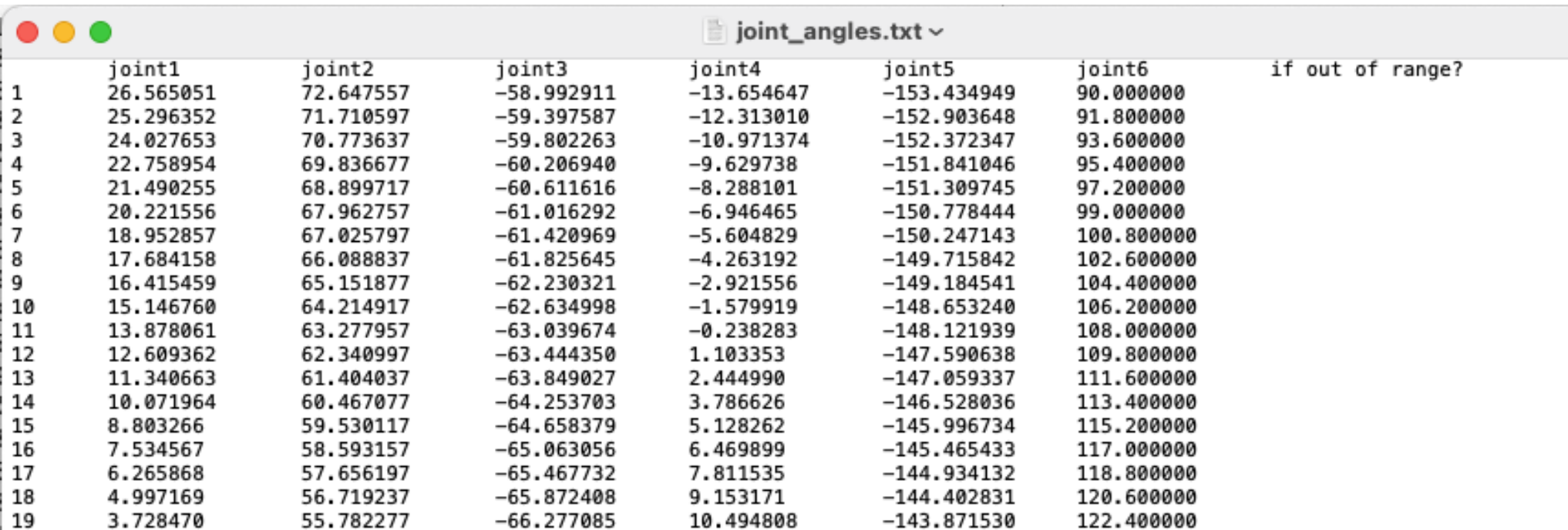


Example :(b) Cartesian move - acceleration



Example : Cartesian move & Joint move dump txt file

```
fileID = fopen('joint_angles.txt', 'w');|
fprintf(fileID, '\tjoint1\t\tjoint2\t\tjoint3\t\tjoint4\t\tjoint5\t\tjoint6\t\tif out of range?\n');
for i = 1:100
    fprintf(fileID, '%d\t', i);
    fprintf(fileID, '%f\t', joint_angle(1:6,i));
    fprintf(fileID, '\n');
end
fclose(fileID); % 關閉檔案
```



	joint1	joint2	joint3	joint4	joint5	joint6	if out of range?
1	26.565051	72.647557	-58.992911	-13.654647	-153.434949	90.000000	
2	25.296352	71.710597	-59.397587	-12.313010	-152.903648	91.800000	
3	24.027653	70.773637	-59.802263	-10.971374	-152.372347	93.600000	
4	22.758954	69.836677	-60.206940	-9.629738	-151.841046	95.400000	
5	21.490255	68.899717	-60.611616	-8.288101	-151.309745	97.200000	
6	20.221556	67.962757	-61.016292	-6.946465	-150.778444	99.000000	
7	18.952857	67.025797	-61.420969	-5.604829	-150.247143	100.800000	
8	17.684158	66.088837	-61.825645	-4.263192	-149.715842	102.600000	
9	16.415459	65.151877	-62.230321	-2.921556	-149.184541	104.400000	
10	15.146760	64.214917	-62.634998	-1.579919	-148.653240	106.200000	
11	13.878061	63.277957	-63.039674	-0.238283	-148.121939	108.000000	
12	12.609362	62.340997	-63.444350	1.103353	-147.590638	109.800000	
13	11.340663	61.404037	-63.849027	2.444990	-147.059337	111.600000	
14	10.071964	60.467077	-64.253703	3.786626	-146.528036	113.400000	
15	8.803266	59.530117	-64.658379	5.128262	-145.996734	115.200000	
16	7.534567	58.593157	-65.063056	6.469899	-145.465433	117.000000	
17	6.265868	57.656197	-65.467732	7.811535	-144.934132	118.800000	
18	4.997169	56.719237	-65.872408	9.153171	-144.402831	120.600000	
19	3.728470	55.782277	-66.277085	10.494808	-143.871530	122.400000	